

• **Answer the following questions in brief :**

(1) What are the effects of human intervention on the environment?

Ans. Most of the effects of human intervention on the environment are of adverse nature.

The adverse effects of human intervention on the environment are as follows :

(1) Climate change (2) Deforestation (3) Loss of Biodiversity (4) Pollution
(5) Ozone depletion (6) Water depletion (7) Habitat Destruction (8) Loss of Wetlands (9) Resource Depletion (10) Genetic modification, etc.

(2) What are the main types of ecosystem? Explain the basis of this classification.

Ans. (A) **Main types** : Land ecosystem and water ecosystem are the main types of ecosystem. (B) **Basis of classification** : Specific locality, climate, topography, water flow system, etc. are factors that act as the basis of classification of ecosystem.

(3) What important role does water play for the environment?

Ans. (1) Water for the environment is in the rivers, streams and also in restoring natural flows linked to wetlands. (2) All living beings are provided with enough water at their essential stages so they don't have shortage of food. Hence they continue their reproduction and growth cycle. (3) Water plays a major role in maintaining the health of rivers and wetland regions. So it supports the ecosystems that depend on them.

(4) Write the ratio of forest cover to total land area in Maharashtra.

Ans. (1) The ratio of forest cover to total land area in Maharashtra is nearly 20.13 %.
(2) Nearly 61935.76 Sq. km land of Maharashtra is covered by forest.

(5) Which is the most forested and least forested region in Maharashtra?

Ans. (A) **The most forested region** : The most forested region in Maharashtra is Gadchiroli district. (B) **The least forested region** : The least forested region in Maharashtra is Latur district.

(6) What are the types of forests found in Maharashtra? Write the percentage of proportion of each type of forest with respect to the area.

Ans. (A) **Types of forests found in Maharashtra** : (1) Evergreen forests (2) Semi-evergreen forests (3) Wet deciduous forests (4) Dry deciduous forests (5) Khajan (Mangrove) forests.

(B) **The percentage of proportion with respect to the area** : (1) Evergreen forests cover nearly 2.54% of total forest area. (2) Semi-evergreen forests cover nearly 7.73% of total forest area. (3) Wet deciduous forests cover nearly 29.84% of total forest area. (4) Dry deciduous forests cover nearly 57.41% of total forest area. (5) Khajan (Mangrove) forests cover nearly 3 % of total forest area.

• **Answer the following questions in brief :**

(1) Explain the relationship between human evolution and the ‘Panchatatvas’.

Ans. (1) Using their intelligence, human beings observed the changes in the nature since ancient times. (2) Human beings came to realize that the origin, operation as well as destruction of all living things as well as non-living things depend on the five major elements found in nature viz, earth, water, fire, air and sky. These five major elements are called ‘Panchatatvas’. (3) Since ancient times, the ideal way of life emerged in India by respecting and faithfully practising the principles of ‘Panchatatvas’.

(2) Write a note on Varahmihira.

Ans. (1) About one and a half thousand years ago in India, Varahmihira wrote four chapters on the subject of water in his treatise (grantha) ‘Brihatsanhita’. (2) It gives the information about soil texture, botanical studies, anthills near trees and also deals with the relation of groundwater to the behaviour of animals such as frog, fish, snake, lizard, mongoose, cobra, turtle, etc.

(3) Write a note on Parashara Muni.

Ans. (1) Indian culture has always been agricultural. (2) Therefore, Parashar Muni had recorded valuable observations in his treatise (grantha) ‘Krisht-Parashar’ to forecast the rainwater required for agriculture.

(4) Give information about ancient lake culture.

Ans. (1) A ‘Lake’ is a natural or artificial reservoir in a shallow area of land. (2) Lakes are given an architecturally determined form by constructing edges, bridges, sculptures, etc. (3) Lakes are also known as Pushkarani, Vapi, Vapika, etc. (4) Lakes and especially Pushkaranis are found in most ancient temple premises in India. It was a practice to build ghats, lamp towers, pillars, big gates, etc. near lakes. (5) Lakes have been used in villages since ancient times for many reasons such as for water supply, adornment of temples, religious activities, as fire extinguisher, for beautification, as facilities for travelers, etc.

(5) Explain the water management during the times of Chhatrapati Shivaji Maharaj.

Ans. (1) Chhatrapati Shivaji Maharaj built many forts in Maharashtra. (2) Most of the forts built by Chhatrapati Shivaji Maharaj have water storage system. (3) Lakes, ponds, etc. were built in most of the forts.

(6) Write in brief about ‘Khandeshi Phad’ Irrigation Method.

Ans. (1) ‘Khandeshi Phad’ irrigation system was built by people of ‘Jire Mali’ community during the 13th to 15th centuries. (2) This method involved the distribution of shared water according to the availability of water to the share of land owned by different persons or families. (3) The river water was diverted towards the field through canal, ‘pat’ by building a causeway of stone. The crop in each field was fixed. (4) The dam was probably built by the rulers and later the canal, ‘char’, ‘pat’ were prepared by beneficiary farmers. (5) Sludge removal and slope repair was done through labour (Shramadan). The guard ‘Patkari’ were from the village.

(7) Write information about ‘Malgujari’ lake.

Ans. (1) During the rule of the king Gaud, the people of ‘Kohli’ tribe used their own money and labour to build about ten thousand cattle ponds in Chandrapur, Gadchiroli, Bhandara and Nagpur districts of Vidarbha. These lakes are known as ‘Malgujari’ lakes. (2) These lakes were built on all four sides of the village connected to each other by canals. (3) There was a beneficiary committee for repair, protection and water distribution. (4) The sludge from the lakes was dumped in the field.

(8) Write information about ‘Pushkarni’.

Ans. (1) The pond that was constructed using the same stone obtained while digging the ground during the construction of temple is called ‘Pushkarni’. (2) Water in ‘Pushkarni’ was used for irrigating temple owned farms, for sacred grove, etc. (3) ‘Pushkarni’ are found near Ajanta, Ellora, Pitalkhore caves in Marathwada region. They are also found near temples in Nashik, Aurangabad, etc.

• **Answer the following questions in brief :**

(1) Explain the V-notch method of measuring flowing water.

Ans. (1) V-notch is a tool made of wooden or iron plank with a V-shaped triangular notch in the centre (with an angle of 90°) to measure flowing water. Depending on more or less water flow, big or small V-shaped notch is used. (2) This wooden plank is placed horizontally in the direction of the flow of water so that the entire stream passes through the notch. (3) The flow of water can be measured in metres per second or litres per second. (4) The flow of water can be measured using a rectangular or trapezoidal notch, but in that case the flow calculation becomes comparatively complicated.

(2) Write units for measuring stagnant water and flowing water.

Ans. (A) **Units for measuring stagnant water :** (1) Litre (2) Cubic metre (3) Hectare-centimetre (4) Hectare-metre, etc.

(B) **Units for measuring flowing water :** (1) TMC (2) Cumec (3) Cusec, etc.

(3) Explain the difference between Cusec and Cumec.

Ans. (1) A unit of flowing water equal to 1 cubic foot per second is called Cusec. On the other hand, a unit of flowing water equal to a cubic metre per second is called Cumec. (2) 1 Cusec is equivalent to 28.317 Litres per second. On the other hand, 1 Cumec is equivalent to 1000 Litre per second.

(4) Write information about evaporation measurement.

Ans. (1) For the measurement of evaporation, a shallow, round iron vessel having 1.2 diameter and 30 cm depth painted white on both sides and placed on a wooden frame for proper aeration from beneath is used. (2) A measuring tape along with an indicator is placed at the centre of the container and the loss of water due to evaporation throughout the day of premeasured water poured into the container is measured at 8 O' clock in the morning every day. (3) Evaporation is measured in inches or millimetres.

(5) How is the water required for crops calculated?

Ans. (1) The total amount of water required for cultivation of the crop till the harvest of crop is defined as the total water requirement of the crop. (2) While measuring the water spread over an area of crop, the depth of the water level in the soil is calculated. It is a commonly used method of measuring water required for crops. (3) Another method of measuring the water requirement for crop is by measuring the per acre of water required for growing the crop.

(6) Write how different flood lines are drawn.

Ans. (A) White Line : After releasing 30,000 cusecs of water from the dam, the flood line is drawn at the level to which the water level rises in the river basin. This line is known as White Line. (2) **Blue Line :** After releasing 60,000 cusecs of water from the dam, the flood line is drawn at the level to which the water level rises in the river basin. This line is known as Blue Line. Once in 20 to 25 years, river water reaches up to Blue Line. (3) **Red Line :** After releasing 100,000 cusecs of water from the dam, the flood line is drawn at the level to which the water level rises in the river basin. This line is known as Red Line. Once in 40 to 50 years, river water reaches up to Red Line due to heavy rainfall.

• **Answer the following questions in brief :**

(1) What is a water plan?

Ans. (1) A plan about storage, measurement, purification and use of water is called a water plan. It is essential to prepare a water plan so that water can be used wisely. (2) While preparing a water plan, various factors like availability of water, exact requirement of water, climate, forest area, number of animals and birds, number of industries, etc. need to be taken into account. (3) Preparation of a water plan and its execution can help in restoring ecosystem (nature) and maintaining the environmental balance.

(2) What are the elements of perfect water plan?

Ans. Some of the elements of perfect water plan are as follows : (1) Collecting information about the available water sources. (2) Gathering information about number of people, animals, birds, industries, land under cultivation, crops grown, etc. in a region. (3) Measuring the amount of rainfall in a region. (4) Measuring groundwater level. (5) Fixing the locations of various water bodies. (6) Finding suitable places for water recharge. (7) Making arrangements for the storage of surface water, well, borewell, etc. (8) Collating information about various water schemes and water available through them and effective implementation of water plan.

(3) What are the steps to be considered while making a water plan?

Ans. (1) Public / General Meeting is called to discuss problems related to the water for finding remedial measures. A water committee is formed in the meeting. (2) Round / Survey Visit is organised to gather information about available water resources, types of forests and soil, possibility of micro and mini water shed areas, etc. (3) All important information related to water is to be recorded in register with reference to a map of a village. (4) The record of total availability of water and requirement of water is maintained. Accordingly, a water plan is prepared and implemented.

(4) What is Public / General Meeting for and who are involved in it?

Ans. (1) The first stage of a water plan is called a Public/General meeting. (2) Children, youth, adults, elders, men and women, etc. are given representation in the Public/General meeting. (3) Public/General Meeting is conducted immediately after the end of the rainy season i.e. in the first week of October. (4) The Public / General Meeting is held in 2 or 3 sessions of about 1-1 hours. In these three sessions, the following three topics are discussed (a) Identified/Expected problems

related to water (b) Current status and (c) Necessary solutions. (5) In a Public / General Meeting, a Water Committee is formed. The selected Water Committee prepares the groundwater plan under the guidance of experts and decides on the implementation framework accordingly.

(5) Who should be included in the survey visit? Why?

Ans. (1) Trained and expert members from Water Committee and experienced and knowledgeable people from village should be included in the survey visit (2) For the proper planning and implementation of the water plan, experienced and knowledgeable people should be included in the survey visit.

(6) What information can you get from the survey visit? Write in detail.

Ans. The information about the following details can be collected from the survey visit : (1) Administrative boundary of the village and Natural boundaries (2) Possibility of establishment of micro and mini watershed areas within the village boundary (3) Water sources available i.e. number of ponds, farm ponds, dams, well, etc. (4) Amount of water required for domestic use, agriculture, industries, animal husbandry, etc. (5) Landforms and geological features of village i.e. texture of soil, slope of land, hilly region, length and width of water bodies like spring, rivers, etc. (6) Type of forests and number of trees, etc.

(7) Use your brain power.

Water plan can be implemented / used even at the household level, identify its components.

Ans. Water plan can be implemented / used even at the household level by considering the following components : (1) To know the source of annual or daily water supply in the house (2) To check whether the water is sufficient or insufficient (3) To check whether the water is clean or dirty (4) To check for what purposes the available water is used (5) To see the water sources available in the area (6) To prepare a water plan based on the amount of rainfall and water storage plan.

(8) Use your brain power.

Identify the types of water wastes and suggest solutions on it.

Ans. (A) Types of water wastes : (1) Habit of brushing or washing dishes by keeping water running. (2) Using much more water than required for washing cars, two wheelers, etc. (3) Wasting water while irrigating plants. (4) Wasting water in industries, offices, schools, etc.

(B) Solutions : (1) Get into the habit of turning on the water tap only when necessary while brushing or washing dishes. (2) Using limited amount of water for washing vehicles and for cleaning purpose. (3) Using drip irrigation or sprinkler irrigation for irrigating plants. (4) Following a water plan for appropriate use of water in industries, offices, schools, etc.

• Answer the following questions in brief :

(1) Why is it necessary to implement watershed development programme?

Ans. (1) Development of watershed area is started from a small watershed area.

(2) Once a small watershed area is selected, its geographical features are studied and a watershed development plan is prepared accordingly. (3) Various treatment methods are used to control soil erosion and to stop the flow of water. (4) Groundwater conservation works are carried out to block and redeem rainwater according to the elevation in a watershed area. Thus, for the development of watershed area, it is necessary to prepare and implement watershed development programme.

(2) Write information about the activities implemented for watershed development in your area.

Ans. (1) *Jaldhara* is a watershed development project implemented in my area. (2) This programme aims at enabling the rural population in my area to conserve water for drinking, irrigation, fisheries and afforestation. (3) Local governing body has taken up construction of various water harvesting structures such as percolation tanks, dug out ponds, check dams, etc. through public participation.

(Note : Students are expected to write answer to this question as per their personal experience.)

(3) What remedial methods are used in development of watershed area?

Ans. The following remedial methods are used in development of watershed area :

(1) Digging Continuous Contour Trenches (2) Building loose boulder structures (3) Nala embankment (4) Building *Vanarai* check dams (5) Building gabion check dams (6) Loop bunding (7) Digging farm ponds, etc.

(4) What can be done to recharge groundwater in urban areas? Explain with an example.

Ans. (1) In urban areas, ground water is pumped through borewells for daily use, factories and industries. (2) For this, it is necessary to refill the tube wells and refill the wells through the government and NGOs. (3) Rainwater harvesting should be done using roofs of buildings. (4) As a result, the runoff rain water can be used.

(5) What is farm pond? How is it constructed?

Ans. (1) Excess running water from the dams is pumped out of the fields by trenches and stored in the shallow areas of field. It is called farm pond. (2) Constructing a farm

pond begins with digging a pit of 30 metres long $\times$ 30 metres wide $\times$ 3 metres deep. (3) The sides and base of the pit is sealed with a proper slope on its sides. (4) About 1500 cubic metres of water can be stored in a farm pond.

(6) How can the technique of 'Block water, Seep water' be made successful while cultivating land? Explain with an example.

Ans. The technique of 'Block water, Seep water' can be made successful while cultivating land by building percolation tank. (2) Percolation tank is an artificially created surface water body to recharge ground water storage. (3) Percolation tanks are designed for storage capacity of 0.1 to 0.5 MCM. (4) The percolation tanks are mostly earthen dams. (5) The purpose of the percolation tanks is to recharge the ground water storage and therefore seepage below the seat of the bed is permissible. (6) Percolation tanks are built on a large scale in many villages of India. (7) The water blocked and seeped into the ground is used for irrigation of crops.

• Answer the following questions in brief :

(1) Why is public participation important in water conservation activity?

Ans. (1) About 75% of rural population in Maharashtra face shortage of drinking water and water for agriculture. (2) Temporary solutions like water supply by tanker are used. However, it is essential to solve the problem of water scarcity permanently through public participation. (3) Large scale public participation makes it possible to allocate various forms of work and to make adequate use of available resources like time, labour, money, etc. and to solve the problem of water scarcity collectively.

(2) Write down the information of an activity of a water conservation which is completed through public participation in your area. Write its objectives.

Ans. (1) Recharge trench is built in my building through public participation for rainwater harvesting. (2) The roof area of my building is nearly 350 sq. m. (3) Members of building built a trench 1 m wide, 1 m deep and 10 m. long. (4) Trench is filled with boulders at the bottom, gravel in between and coarse sand at the top so that slit from runoff can be removed easily. (5) A mesh was provided at the roof to prevent leaves and other waste to enter the trench. (6) By pass arrangement is provided before the collection of rainwater to reject the first showers. (7) The stored water in a trench is used for irrigation of plants in building, washing vehicles, cleaning the premises of building, etc.

(Note : Students are expected to write answer to this question as per their personal experience.)

(3) What are the objectives of public participation?

Ans. The following are the objectives of public participation : (1) To preserve the natural sources of water (2) To solve the problem of water scarcity in a region (3) To provide permanent solutions to problems related to water (4) To use available resources like time, labour, money, etc. for the wellbeing of all.

(4) Use Your Brain Power

Mention the characteristics of public participation in the following diagram.

Ans. The characteristics of public participation are as follows : (1) More work in less time (2) Division of work and responsibility (3) Reduced cost and economic development (4) Harmony and peace in village (5) Problem solving (6) Skill development.

• **Answer the following questions in brief :**

(1) What type of wind causes rain in Maharashtra?

Ans. (1) Maharashtra state receives the maximum rainfall of 85 % from the Southwest monsoon winds from June to September. (2) From the end of September and throughout October, monsoon winds become weak and start to retreat. Therefore the month of October is called 'Transitional Period'. (3) From the end of October till January, Northeast monsoon winds bring occasional retreating rains in Maharashtra.

(2) What is a drought-prone region?

Ans. An area on the leeward (downwind) side of a mountain where precipitation is significantly less than that on the windward (upwind) side is considered to be rain shadow region or drought-prone region.

(3) How is the distribution of water in Maharashtra? Write in detail.

Ans. (1) The Maharashtra state receives rains of an average of 1360 mm annually. However, the distribution of rainfall in the state is uneven. (2) The Konkan region located to the west of Sahyadri mountains receives an average of 2000 mm to 3500 mm rain annually. (3) However, the area to east of Sahyadri mountains i.e., some area of Plateau region (drought-prone region) receives an average of 300 mm to 500 mm rain only. (4) The eastern part of Maharashtra receives an average of 1000 mm to 1400 mm rain annually. (5) About 55% of the total rainfall in Maharashtra falls in Konkan. The rest of Maharashtra receives about 45% rainfall.

(4) What are the remedial measures to deal with current water crisis in Maharashtra?

Ans. The remedial measures to deal with current water crisis in Maharashtra are as follows : (1) Arranging extensive public awareness programmes (2) Adopting ground water recharge measures (3) Purifying polluted water (4) Using drip irrigation, sprinkler irrigation and micro irrigation (5) Removing sludge from lakes and dams (6) Constructing Besalt / Pashan lakes (7) Making everyone accustomed to water audits regarding domestic water use or industrial water use (8) Arranging various training sessions for increasing water literacy, etc.

(5) What are the causes of drought-prone conditions in Maharashtra? Suggest remedies for it.

Ans. (A) **Causes of drought-prone conditions in Maharashtra :** (1) Unequal distribution of water resources (2) Lack of effective planning (3) Neglect of ground water

(4) Increasing demand of water for civilization and industrialization (5) Inadequate utilization of irrigation capacity (6) Water pollution, etc.

(B) Remedies to overcome drought-prone conditions in Maharashtra :

(1) Adopting ground water recharge measures (2) Using drip irrigation, sprinkler irrigation and micro irrigation (3) Constructing Besalt / Pashan lakes, etc.

(6) Nowadays water management has become very essential. Write your opinion about the same.

Ans. (1) Due to increasing population, the demand for water in India has increased to a great extent. (2) Many villages are facing drought conditions. (3) Due to irregular and insufficient rainfall, agricultural production has decreased in many villages in India. (4) Scarcity of water has created health related issues in some villages of India. (5) At many places, surface water or groundwater is wasted, water is overused or water is polluted. Therefore, nowadays water management has become very essential for the revival of traditional water bodies, construction of water storage infrastructure, and safe disposal of wastewater.

• **Answer the following questions in brief :**

(1) What is irrigation? What are the main objectives of irrigation?

Ans. (A) Meaning of irrigation : The process of giving additional water to crops other than rainwater is called irrigation.

(B) Main objectives of irrigation : (1) To make water available for taking crops in seasons other than rainy season. (2) Successfully harvesting more than one crop in a year. (3) To make adequate water available for cash crops (4) To get higher yield per hectare through proper irrigation. (5) To do protected farming by providing assured and sufficient water, etc.

(2) What are the main types of irrigation?

Ans. The main types of irrigation are as follows : (1) **Watering from the surface of the land :** In this type of irrigation, furrow and bed types of irrigation are used.

(2) **Spraying water from certain height :** In this method of irrigation, water is sprayed on crops from certain height. (3) **Watering near the roots of plants :** In this method of irrigation, water is given to the crops near their roots.

(3) Drip irrigation is mainly used for which crops in Maharashtra? Why?

Ans. (1) Drip irrigation is mainly used for crops like sugarcane, cotton, grapes, banana, oranges, rose, gerbera, coconut, oil seed, etc. in Maharashtra. (2) Drip irrigation is used in Maharashtra due to the following reasons : (a) plants receives water as per the exact requirement (b) 30% to 80% water is saved (c) soil erosion is controlled (d) 30% to 35% expenses are saved, etc.

(4) What are the components of drip irrigation set? What are the benefits of the method of drip irrigation?

Ans. (1) Pipes, valves, ploythene tubes and emitters are the components of drip irrigation. (2) The benefits of the method of drip irrigation are as follows : (a) plants receive water as per the exact requirement (b) 30% to 80% water is saved (c) soil erosion is controlled (d) 30% to 35% expenses are saved (e) suitable for saline soil, uneven land (f) water is conserved, etc.

(5) Compare drip irrigation and sprinkler irrigation method.

Ans. (1) In drip irrigation, the water is given to the crops around their roots. On the other hand, in sprinkler irrigation, the water is sprayed on the crops from certain heights. (2) Comparatively less area is wetted by drip irrigation. On the other hand, comparatively more area is wetted by sprinkler irrigation. (3) Run off and evaporation is comparatively less in drip irrigation. On the other hand, run off and evaporation is comparatively higher in sprinkler irrigation.

(6) What kind of material is used for sprinkler irrigation?

Ans. The following material is used for sprinkler irrigation : (1) Polyvinyl (PVC) pipe or polyethylene pipe (2) Control valves (3) Risers (4) Sprinkler heads (5) Automatic timer (6) Pipe cutter, etc.

• **Answer the following questions in brief :**

(1) Write information on International Corporation Agreement between India and Bangladesh.

Ans. (1) International cooperation agreement was signed between India and Bangladesh in 1972. (2) In this agreement, the following points related to water resources were agreed upon by both the countries : (a) To study water use in Ganga Project Plan. (2) To provide flood forecasting information about rivers like Ganga, Brahmaputra, Barak, etc., by India to Bangladesh.

(2) Which topics are covered in the government policy of irrigation?

Ans. The following topics are covered in the government policy of irrigation : (1) Government policies on irrigation are related to adequate storage and efficient use of water resources. (2) Government policies on irrigation stress upon the sub-basin wise planning of water management of important rivers and watershed areas. (3) In times of crisis like drought, food shortage, etc., the government also passes orders indicating changes in the use of water.

(3) Write objectives of water planning in Maharashtra.

Ans. Objectives of water planning in Maharashtra are as follows : (1) Basin and sub-basin wise planning with respect to five main river basins. (2) Decentralization of management of water resources for the development of catchment areas. (3) Establishment of Water Resources Development Corporation for river basin wise utilisation of water resources and overall development. (4) Integration of all types of water management plans. (5) Encouraging local bodies and people through public participation. (6) Implementation of schemes for improving water quality management and its long term maintenance.

(4) What are the objectives or policies of the Irrigation Department?

Ans. The objectives or policies of the Irrigation Department are as follows : (1) To plan policies for the water conservation (2) To ensure enough moisture essential for plant growth (3) To provide crop insurance against short duration drought (4) Encouraging women's participation in water management programme, etc.

(5) Compliance with water laws and regulations is important for every nation.

Write your opinion on this.

Ans. (1) Rising demand for water due to rising population is likely to lead to local, national or international disputes. Therefore, water acts and water rules are highly essential. (2) Many countries in the world are in conflict over the distribution of river water. International cooperation agreements have been signed between many countries to resolve these disputes. Thus, compliance with water laws and regulation is important for every nation.

CHAPTER 1

WATER PURIFICATION AND SEWAGE (WASTEWATER) MANAGEMENT

• Answer the following questions in brief :

(1) What is the difference between drinking water and river water?

Ans. (1) Drinking water is relatively purer. On the other hand, river water is relatively less clean. (2) Drinking water is mainly used for domestic purposes. On the other hand, river water is used for domestic consumption, agriculture, factories, power generation, etc. (3) Drinking water is relatively less available. On the other hand, river water is relatively more available.

(2) Why is it said that untreated water is not suitable for drinking?

Ans. (1) Untreated water and poor sanitation are linked to transmission of diseases such as cholera, diarrhoea, dysentery, hepatitis A, typhoid and polio. (2) Untreated water may be of poor quality related to its taste, smell and visual appearance.

(3) Which water purifier tools are used in houses? Describe them.

Ans. (1) Ultraviolet radiation water purifier, Iron exchange filters, ozone gas water purifiers, tap filters, life straws, candle filters, etc. are used as water purifier tools in houses. (2) Most of these tools use activated carbon, polymer resin, etc. for removing dust and fine sand particles from water. (3) Most of these tools help in killing the germs and viruses present in water. (4) The tools like life straws are convenient to carry while travelling. (5) Tools like candle filters are used if water is less contaminated. Candle filter contains vessels placed one above the other. Filter rods are placed at the bottom of the upper vessel. The water in the upper vessel is filtered through rods and is stored in the lower vessel.

(4) What are the important stages implemented in water purification?

Ans. (1) Water is first pumped from a water source. Water is pumped out using electricity pumps. (2) In the stage of aeration, air is circulated through, mixed with or dissolved in the water. (3) After aeration, coagulation is done to remove the fine particles suspended in the water. (4) In the stage of flocculation, water is gently stirred by low speed mixing blades in a number of compartments of flocculation tank for nearly thirty minutes. (5) In the stage of sedimentation, the water is settled down. (6) In the process of filtration, fine particles, viruses, bacteria, etc. are removed from water. (6) In the stage of chlorination, chlorine is added to water. (7) At last purified water is stored in large tanks and distributed to various regions through pipelines.

(5) How is water purification process done for big cities / villages? Write briefly.

Ans. (1) Rainwater or flowing water is stored and then is pumped to water treatment area. (2) In the stage of aeration, air is circulated through, mixed with or dissolved in the water to remove undesirable taste and odour from the water. (3) After aeration, coagulation is done to remove the fine particles suspended in the water. (4) In the stage of flocculation, water is gently stirred by low speed mixing blades in a number of compartments of flocculation tank for nearly thirty minutes. (5) In the stage of sedimentation, the water is settled down. (6) In the process of filtration, fine particles, viruses, bacteria, etc. are removed from water. (6) In the stage of chlorination, chlorine gas or disinfectants like ozone, bromine, etc. are added to water. (7) At last purified water is stored in large tanks and distributed to various regions at high pressure using channels or pipelines.

(6) What are the advantages and disadvantages of purifying water with UV rays?

Ans. (A) **Advantages :** (1) It eliminates nearly 99.9% of microorganisms. (2) It does not affect the colour or the taste of water. (3) It uses no chemicals for purifying water. (B) **Disadvantages :** (1) Though it kills germs and viruses, they are not removed from the water. (2) These devices cannot be used without power supply.

• **Answer the following questions in brief :**

(1) What is Earth Overshoot Day?

Ans. (1) Nature gives us resources in many different ways. (2) Due to population explosion, industrialisation, etc. natural resources are being used extensively in modern times. (3) As a result, in modern times, the resources in nature's banks are dwindling. This condition is called Earth Overshoot Day. (4) Earth's Overshoot Day is calculated illustrative date on which humanity's resource consumption for the year exceeds the Earth's capacity to regenerate those resources that year.

(2) What are the incidences and factors involved in polluting water in daily life?

Ans. The incidences and factors involved in polluting water in daily life are as follows :
(1) Cleaning routine (2) Shaving (3) Taking bath (4) Washing clothes (5) Washing utensils (6) Toilet cleaning (7) Using cosmetics (8) Carrying out rituals, etc.

(3) How to reuse wastewater?

Ans. Wastewater can be reused as follows : (1) Reusing wastewater for irrigation of crops (2) Reusing wastewater for cleaning premises of buildings (3) Reusing wastewater for cleaning vehicles (4) Reusing waste water for electricity generation (5) Reusing waste water for textile industries, etc.

(4) Make rules about the precautions to be taken by house members to maintain water quality.

Ans. (1) Storing water in a clean place. (2) Keeping water storage containers clean. (3) Covering drinking water (4) Filtering and boiling drinking water (5) Spinning alum in drinking water to purify it, etc.

(5) Explain how water is wasted.

Ans. (1) Using excessive water for washing vehicles (2) Keeping the water tap running while washing clothes or cleaning utensils. (3) Keeping the water tap running while brushing or bathing. (4) Wasting water while watering plants (5) Wasting water while watering plants in public crops or crops in the fields. (6) Wasting water by overusing water for cleaning routines, etc.

(6) How sugar mills fulfill their water needs?

Ans. (1) As per Water Prevention and Control of Pollution Cess Rules 1978, the water requirement for sugar industry is 2.0 m³/ton of cane crushed and wastewater generation as 0.40 m³/ton of cane crushed. (2) It takes about 210 litres of water to produce 1 kg of sugar cane. (3) Water is sprayed on the sugarcane to achieve maximum juice extraction while it is being crushed.

(7) What happens to sewage water after it is treated?

Ans. (1) Sewage water becomes usable for gardening, cleaning vehicles, cleaning toilets, cooling equipment, irrigation, etc. after it is treated. (2) Sewage water can be reused for drinking if it is purified up to 100%.

• **Answer the following questions in brief :**

(1) Why is it said that saline sea water is also important?

Ans. (1) Nearly 97% of the water on the Earth is saline sea water. Only 3% of the water on the Earth is fresh water. (2) World population is increasing day by day. Therefore, world is facing a problem of shortage of fresh water. (3) Therefore, desalination of this brackish water becomes essential to tackle the problem of shortage of fresh water. Therefore, it is said that saline sea water is also important.

(2) Write in brief about the financial affordability aspect of desalination of saline water.

Ans. (1) The average cost of converting sea water into desalinated water is about only 10 paise per litre. (2) This cost can be further reduced if unconventional energy resources are used. (3) The Ministry of Urban Development of the Government of India plans to build 100 smart cities near the coastline which will have desalination centres. These centres' cost of desalination of water will be around 4 to 5 paise per litre.

(3) How groundwater would have become saline in the saline water belt in Maharashtra?

Ans. (1) According to geologists, the salty water of sea from the west seeped into the eroded land of Amravati, Akola, Buldhana, etc. and settled there. (2) Eventually sand and soil accumulated there and as a result layers of yellow soil and sand were formed. (3) Due to these layers, the intruding water of the sea i.e. saline water was trapped there beneath the surface water. In this way, groundwater became saline in the saline water belt in Maharashtra.

(4) What efforts will have to be made to ensure that groundwater in the saline belt does not become saline?

Ans. (1) Without touching the river valleys in the saline belt of Vidarbha, the rest of the streams and nalas can be extended up to 1000 metres in length, deepened up to 20 metres and widened up to 30 metres. (2) Constructing cement dam in nearby region for blocking and storing about 6 lakh cubic metres of fresh water flowing from expanded streams and nalas.